Name: ________________________ **Date:** ______________

Part A: Beginning: Linear Equations and Systems Fundamentals

1. A) **[MC]** What is the primary goal when solving a linear equation for a variable?
   B) To make the equation more complex
   C) To isolate the variable on one side of the equation
   D) To graph the equation
   E) To find the sum of all numbers in the equation

label=E)

   F) $x^2 + 3 = 7$
   G) $5x - 2 = 13$
   H) $\frac{1}{x} = 4$
   I) $xy = 10$

label=I)

Part B: Beginning: Linear Equations and Systems Fundamentals

(j) One

(k) Zero

(l) Infinite

label=L)

(m) Challenge: Liam

(n) *(Extra Practice)* Alice

(o) *(Extra Practice)* Liam

(p) Solve: $6x = 42$. $x =$ ___________

(q) Solve: $\frac{y}{7} = 5$. $y =$ ___________

(r) Solve: $-4a = 32$. $a =$ ___________

(s) Solve: $2m + 9 = 25$. $m =$ ___________

(t) Solve: $5p - 3 = 27$. $p =$ ___________

Part C: Beginning: Linear Equations and Systems Fundamentals

(u) Solve: $18 - k = 10$. $k =$ ___________

(v) Solve: $\frac{x}{2} + 6 = 14$. $x =$ ___________

(w) Solve: $12 = 3 + 3n$. $n =$ ___________

(x) **Word Problem:** A number decreased by 15 is 40. Write an equation and solve for the number. Equation: ________________________________Number: ___________

(y) **Word Problem:** Four times a number is 52. Write an equation and solve for the number. Equation: ________________________________Number: ___________

(z) Distribute and simplify: $5(x+7)$. Answer: ______________________________

() Distribute and simplify: $-3(2y-4)$. Answer: ______________________________

() Combine like terms: $7a+12-3a-5$. Answer: ______________________________

() Combine like terms: $9m-2+m+8$. Answer: ______________________________

() Solve: $3(x-2) = 15$. $x =$ __________

() Solve: $6y + 10 - 2y = 30$. $y =$ __________

() Solve: $4p + 7 = p + 16$. $p =$ ____________

Part D: Beginning: Linear Equations and Systems Fundamentals

() Solve: $2(k+5)-3=19$. $k=$ ___________

() Solve: $\frac{3x}{4}=12$. $x=$ ___________

() Solve: $\frac{z-3}{5}=4$. $z=$ ___________

() Solve: $5x-8=5x+2$. Number of solutions: ___________

() Solve: $4(y+1)=4y+4$. Number of solutions: ___________

A) **[MC]** What does it mean for a linear equation to have "no solution"?
B) The variable equals zero.

C) There is no value for the variable that makes the equation true.
D) Any value for the variable makes the equation true.
E) The equation is too difficult to solve.

label=E)

F) Only one specific value makes the equation true.
G) No value for the variable makes the equation true.
H) Any real number value for the variable makes the equation true.
I) The equation has many variables.